Assignment 1

MSDS 458

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Abstract

This assignment explores how hidden-layer neurons in a single-hidden-layer dense neural network learn the internal representations for handwritten digit classification. Using the MNIST dataset, the main objective is to interpret what hidden nodes respond to as the model capacity increases. A total of 10 experiments were completed while using fully connected networks trained via backpropagation, beginning with extremely low-capacity architectures, which would be about one or two hidden neurons. As experiments continue, the hidden layers grow in size (3, 5, 20, 50, 85). In order to evaluate the behavior of models, accuracy curves, confusion matrices, inspecting misclassified digits, and analyzing scatterplots of hidden-layer activations were helpful. This experimental approach equips both a performance comparison and an interpretive perspective into how internal feature spaces develop as neural networks become wider.

The results indicate that expanding the hidden layer substantially improves the network's ability to separate digit classes within its learned feature space. Accuracy advances from harsh underfitting with only one hidden neuron (approximately 38 percent) to strong generalization at moderate hidden-layer widths (about 90 percent with five neurons). Past this point, adding more hidden units leads to only small gains in accuracy. Performance increases to about 97 percent with 85 neurons, which shows that the benefits of extra capacity begin to level off. These findings demonstrate an obvious transition from representational bottlenecks in small models to more distributed encodings in higher-capacity networks.

More experiments using PCA and Random Forest feature selection imply that while simplified input representations can still grasp some important information, compressing the data too much reduces accuracy by removing pivotal shape and structural details. Overall, the results show how hidden neurons progress from detecting simple stroke-level patterns in very small

models to learning more distributed, high-dimensional representations as the network grows wider. At the same time, these findings highlight the limitations of fully connected dense networks for image data and help explain why convolutional neural networks, which take advantage of spatial structure, are a better fit for achieving stronger performance. Ultimately, this work reiterates that model accuracy and interpretability are closely tied to representational capacity, and that architectural design choices strongly influence what neural networks are able to learn from image data.

Introduction

Handwritten digit classification is an elementary concern in machine learning and is still very relevant in business and operational settings today. Organizations depend on systems to recognize handwritten forms, interpret bank checks, read postal codes, digitize medical records, and extract numeric information from customer documents. There are several more examples, but in these environments, classification accuracy is very important because errors can lead to inefficiencies, incorrect transactions, or reduced customer trust. Even small error rates can scale into large operational costs when digit recognition systems are deployed at high volume. Because of this, digit classification is such a useful example of how machine learning models decipher directly into business outcomes.

Although neural networks gain strong predictive performance on image classification tasks, high accuracy singled out does not explain how these models will make decisions internally. In business contexts where transparency and accountability matter, it is very important to understand what features neural networks learn and why specific classifications occur in general, at least for applicable deployment. This definitely motivates an investigation into the

role of hidden-layer neurons, which serve as internal feature detectors that transform pixel inputs into digit predictions. Hidden layers are frequently described as the “black box” portion of neural networks, yet they are where the most important feature learning occurs. Studying these internal activations helps connect model performance with the representations that make classification possible.

The purpose of this research is to explore how representational capacity emerges in a simple and fully connected neural network with a single hidden layer trained via backpropagation. Instead of only focusing on maximizing accuracy, the prominent objective is interpretability: identifying what the hidden nodes respond to and how class separability evolves as hidden-layer width increases. Using the MNIST dataset, this study evaluates model performance, confusion matrices, activation patterns, and the impact of dimensionality reduction via PCA and feature selection. By gradually growing model capacity, this work provides an unambiguous experimental view of how neural networks shift from underfitting to effective generalization. Ultimately, the assignment underscores the relationships among architectural size, the learned feature space structure, and the practical limits of dense networks for image-based duties.

Literature Review

Neural network training is fundamentally based on backpropagation, which was first formalized by Rumelhart, Hinton, and Williams. They enabled multilayer networks to efficiently learn internal representations through gradient-based optimization (Rumelhart, Hinton, and Williams 1986). This breakthrough made it possible for neural networks to move past simple linear models and begin learning hierarchical feature transformations. Backpropagation remains

the foundational algorithm behind most modern deep learning systems, including the dense architectures explored in this study. Because it adjusts weights layer by layer based on error signals, backpropagation lets hidden neurons gradually specialize in useful patterns during training. This process is what helps neural networks learn features automatically rather than counting on rules that are hand-designed.

Handwritten digit recognition research often relies on the MNIST dataset, which was originally introduced by LeCun and his collaborators (LeCun, Cortes, and Burges 1998). MNIST is widely used because it provides a standard benchmark of 60,000 training images and 10,000 test images, which ultimately makes it a strong dataset for studying how changes in neural network architecture affect classification performance. Because the dataset is simple but still has real handwriting variation, it is superior for analyzing learning behavior, confusion patterns, and representational limits. MNIST has therefore become one of the most common introductory datasets for evaluating neural network classifiers. In particular, MNIST is useful for controlled experiments in which model capacity can be adjusted and the effects on accuracy and internal feature structure can be clearly observed. Its widespread adoption also enables results to be directly compared with earlier work in the neural network literature.

Outside of accuracy, contemporary research emphasizes the interpretability of hidden neurons. Olah, Mordvintsev, and Schubert present feature visualization methods that reveal the specific stimuli to which internal units respond within image classifiers (Olah, Mordvintsev, and Schubert 2017). This highlight on interpretability directly motivates the present analysis of hidden-layer activations and neuron response patterns. Interpretability is especially important when models are applied in real-life environments, where understanding why a prediction occurs can matter as much as the prediction itself. Feature visualization supplies a valuable framework

for connecting internal activations to compelling visual structures. This line of research shows that neural networks are not just memorizing answers, but are actually building layered representations that recognize edges, textures, and eventually more complex shapes. Even in smaller dense networks, hidden neurons can still be seen as early feature detectors that respond to simple stroke characteristics such as curvature or thickness. Because of this, interpretability methods provide valuable context for understanding how hidden-layer width influences what the network learns internally.

Methods

This research used a controlled experimental design to study how hidden-layer neurons in a single-hidden-layer dense neural network learn internal representations for handwritten digit classification. All experiments were performed on the MNIST dataset of 70,000 grayscale digit images. Each 28×28 image was flattened into a 784-dimensional vector and normalized to the range [0,1]. The dataset was split into 60,000 training images and 10,000 test images, with 5,000 training samples held out for validation.

All models were fully connected neural networks trained via backpropagation. Each network contained one hidden layer with ReLU activation and a 10-class softmax output layer. The main experimental variable was hidden-layer width, which was systematically varied across architectures with 1, 2, 3, 5, 20, 50, and 85 hidden units.

After evaluating the models through training, validation, and test accuracy, and loss curves, confusion matrices were calculated to specify structured digit confusions. Interpretability was highlighted through analysis of hidden-layer activations using boxplots and scatterplots, as well as visual inspection of misclassified digit examples. In addition to overall accuracy, these

visualizations were very helpful in showing what the network learned internally. For example, the confusion matrices helped show which digit pairs were still difficult even as the capacity increased, while activation scatterplots provided a view of how separable the learning feature space became as more hidden neurons were added throughout testing. These evaluation methods let the experiments assess how hidden-layer representations changed across architectures.

Two-dimensional reduction experiments were also performed, which was interesting to observe. PCA decreased the input space from 784 pixels to 154 components, which caught 95 percent of the variance, and the best-performing architecture (Hidden = 85) was retrained on this compressed representation. Random Forest feature importance was used to select the top 70 most informative pixels, and the neural network was retrained using only these reduced features to evaluate the impact of aggressive feature selection.

Overall, this methodology combines systematic architectural variation, quantitative evaluation, and activation-based interpretability techniques to understand how hidden neurons evolve as network capacity increases.

Results

The results show that hidden-layer dimensionality enormously determines both classification performance and the structure of learned representations. In Experiment 1, the single-neuron network severely underfit MNIST, which achieved only 38 percent accuracy. Predictions collapsed onto a few prevailing classes, and activation boxplots showed a heavy amount of overlap across digits, which confirms that one hidden neuron can only grasp a crass projection. This experiment operated as an important baseline, which showed that extremely low-capacity networks cannot form significant internal representations for ten-class digit

classification. The results also demonstrate how hidden-layer dimensionality directly limits how well digit classes can be separated in feature space. With only one hidden unit, the network is essentially forced to treat many different digits as visually similar, as it lacks sufficient internal dimensions to encode meaningful differences.

Experiment 2 exhibited that increasing the hidden layer to two neurons dramatically improved performance to 71 percent accuracy, which was interesting to see. All digit classes were predicted, and scatterplots revealed partial geometric separation in a two-dimensional embedding. However, systematic confusions persisted between visually similar digits like the numbers 5 and 8. This means that even small increases in hidden-layer width let the model to learn more structured representations, but low-dimensional feature spaces still struggle with topologically similar digits. These results show that the network begins to develop a more useful internal geometry, yet still lacks enough complexity to fully separate all ten digit classes.

Experiments with three and five hidden neurons produced the most significant transition in performance. Accuracy rose to 82 percent at three neurons and nearly 90 percent at five neurons. Confusion matrices became increasingly diagonal, and hidden activation clusters tightened, suggesting that additional neurons began specializing in complementary stroke features. At this stage, the model is no longer collapsing predictions and instead is learning more distinct feature detectors that support stronger generalization.

After about five hidden neurons, the performance gains started to stabilize. The model with 20 hidden units reached approximately 95 percent accuracy, and the mistakes that remained mostly transpired between visually similar digit pairs like 4 and 9 or 5 and 8. In Experiment 3, increasing the hidden layer further to 50 and 85 units produced only small additional improvements (around 96–97 percent), while signs of overfitting became more noticeable. At

these larger sizes, the hidden representations also became more spread out and harder to interpret using simple low-dimensional visualizations. This mirrors an evident diminishing-returns effect, where additional capacity improves training performance more than it improves generalization. At higher widths, digit identity becomes encoded across many neurons rather than through a small number of interpretable features. This means that once the network has enough neurons to capture most digit structure, further expansion mainly adds redundancy rather than major new discriminative power.

Experiment 4 showed that reducing the input space using PCA still preserved much of the important information needed for classification, but the Hidden = 85 model reached its best validation performance early in training. This shows how important it is to choose models based on validation results rather than simply training longer. Experiment 5 found that restricting the network to only the top 70 most important pixels caused accuracy to drop significantly (to about 92.9 percent), suggesting that digit recognition depends on having a broader spatial context, not just the most obvious stroke pixels. These experiments reinforce that both model capacity and input richness matter for keeping strong digit separability.

Overall, accuracy improves vastly as the hidden layer grows to a reasonable size, but then begins to balance as the network becomes wider. At higher capacities, most remaining errors are due to natural ambiguity in handwriting rather than major limitations in the model itself. This spotlights an important conclusion of the study: improvements in neural network performance are rapid early on, but eventually become constrained by dataset ambiguity and architectural limits.

Conclusions

This research shows that the most important factor affecting both performance and interpretability in a single-hidden-layer dense neural network is the size of the hidden representation. When the hidden layer is extremely small, the model is forced to compress digit images into a very limited feature space, which leads to severe underfitting and predictions collapsing onto only a few classes. As the hidden layer becomes wider, the network moves from detecting only simple stroke patterns to developing more organized internal feature spaces, where digit classes become increasingly easier to separate.

Moderate hidden-layer sizes appear to propose the best balance between accuracy, generalization, and interpretability. Accuracy improves sharply from one to five hidden neurons, which reaches nearly 90 percent, while larger increases begin to show diminishing returns. Networks with 20 to 85 hidden units achieve high performance around 96–97 percent, but the improvements become small, overfitting becomes more noticeable, and the learned representations grow more distributed and harder to interpret using simple low-dimensional visualizations.

The dimensionality reduction experiments further show that accurate classification depends on keeping enough structural information from the original images. PCA was able to preserve much of the predictive signal, but Random Forest feature selection showed that limiting the input to only the most “important” pixels removes rudimentary shape and context. As a result, effective digit recognition relies on broader, distributed spatial information across the entire image rather than a small subset of features.

From a deployment standpoint, organizations should not choose models based only on having the largest network or the highest training accuracy. Instead, model selection should focus

on validation performance, stable generalization, and how understandable the learned features are. In addition, the performance limits observed with dense networks make a case for the value of convolutional architectures, which take advantage of spatial structure and weight sharing to achieve near-perfect digit recognition (LeCun, Bengio, and Hinton 2015).

In conclusion, this study provides experimental evidence of how hidden neurons evolve from uncultivated stroke detectors to distributed feature encoders as capacity increases, which ultimately motivates convolutional approaches for concrete handwritten digit classification.

References

LeCun, Yann, Yoshua Bengio, and Geoffrey Hinton. “Deep Learning.” *Nature* 521 (2015): 436–444. https://www.nature.com/articles/nature14539.

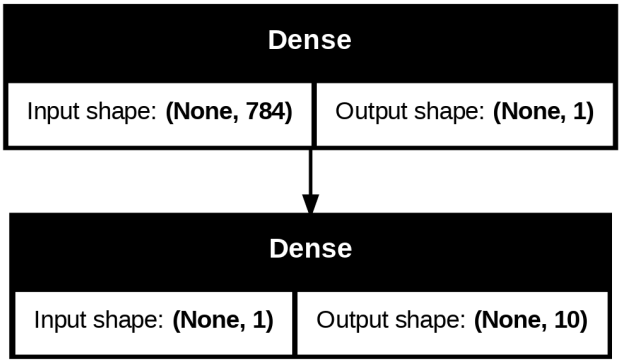
LeCun, Yann, Corinna Cortes, and Christopher J. C. Burges. “The MNIST Database of Handwritten Digits.” 1998. http://yann.lecun.com/exdb/mnist/.

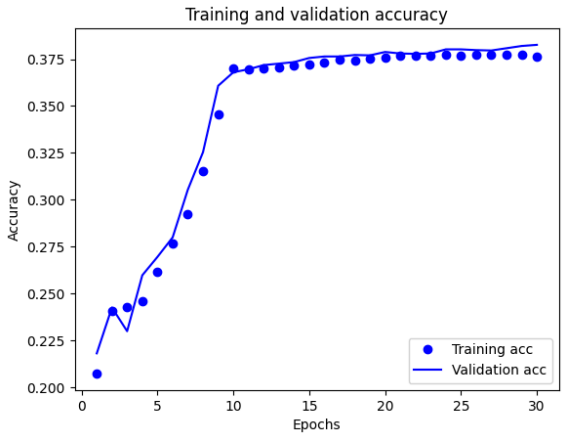
Olah, Chris, Alexander Mordvintsev, and Ludwig Schubert. “Feature Visualization.” *Distill*, November 7, 2017. https://distill.pub/2017/feature-visualization/.

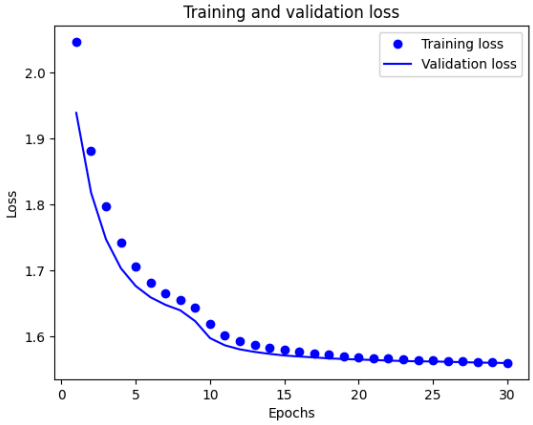
Rumelhart, David E., Geoffrey E. Hinton, and Ronald J. Williams. “Learning Representations by Back-Propagating Errors.” *Nature* 323 (1986): 533–536. https://www.nature.com/articles/323533a0.

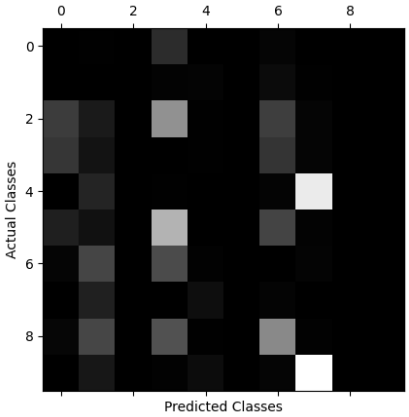
Appendix

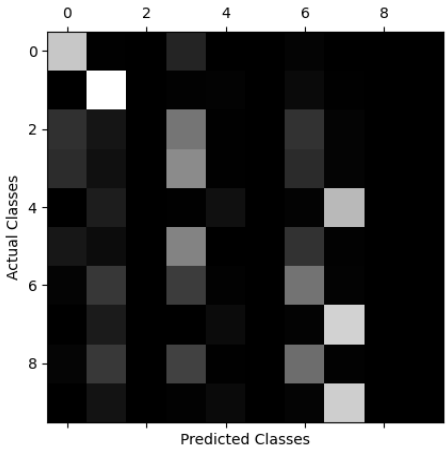
Experiment One

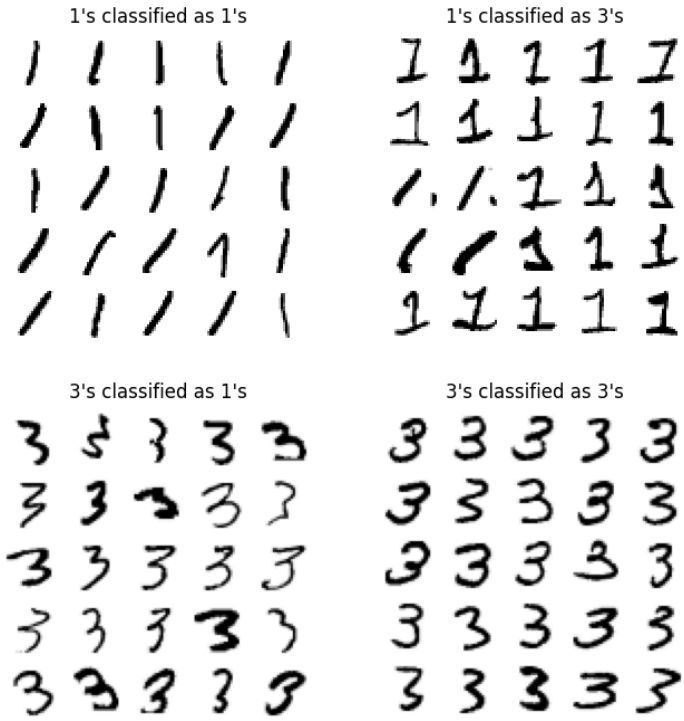




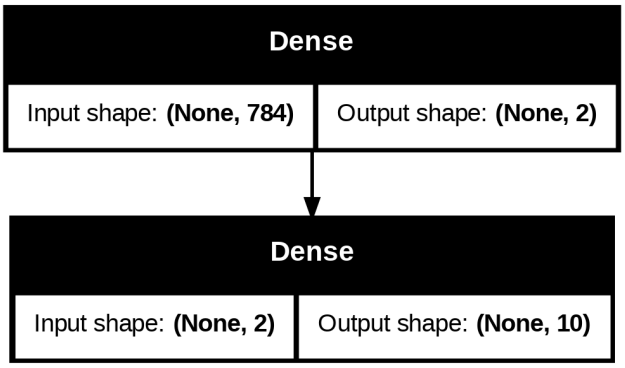


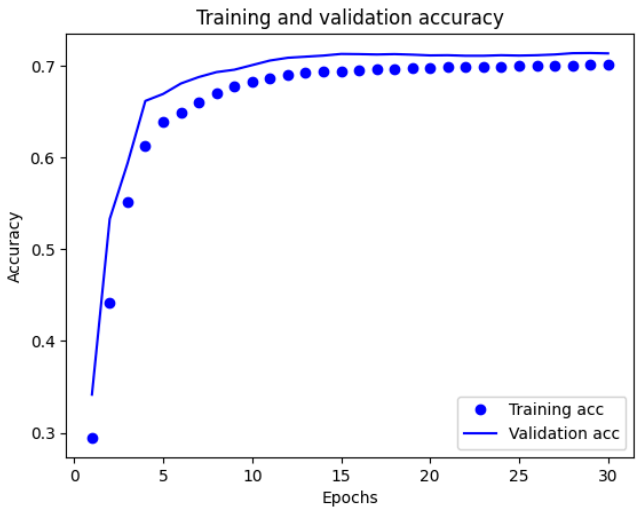






Experiment 2


Dense
Input shape: (None, 784)
Output shape: (None, 2)
Dense
Input shape: (None, 2)
Output shape: (None, 10)


Training and validation accuracy
Accuracy
Epochs
Training acc
Validation acc
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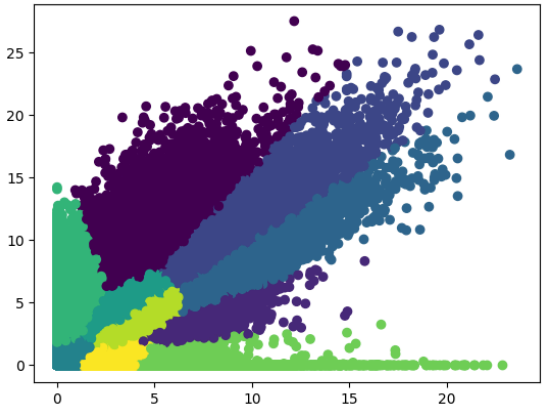
0.6
0.5
0.4
0.3
0
5
10
15
20
25
30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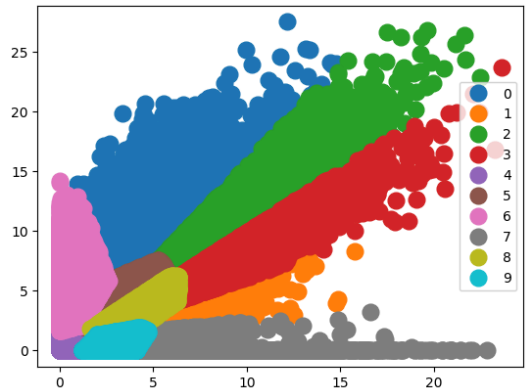
20



Experiment 3.1

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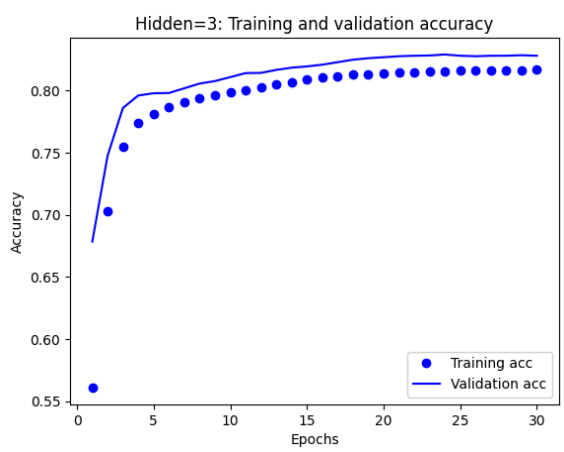
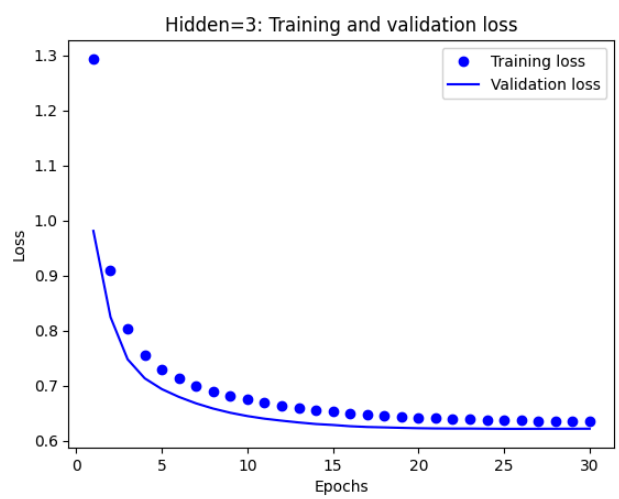
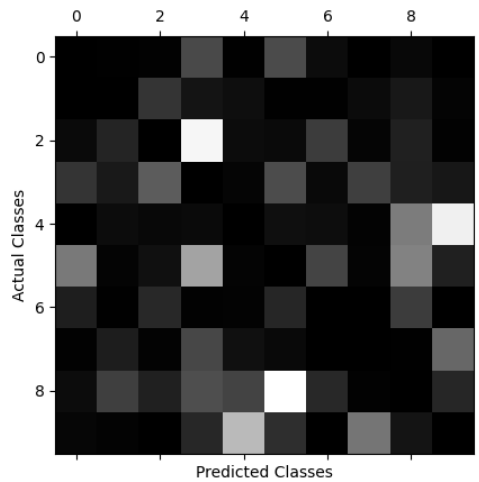
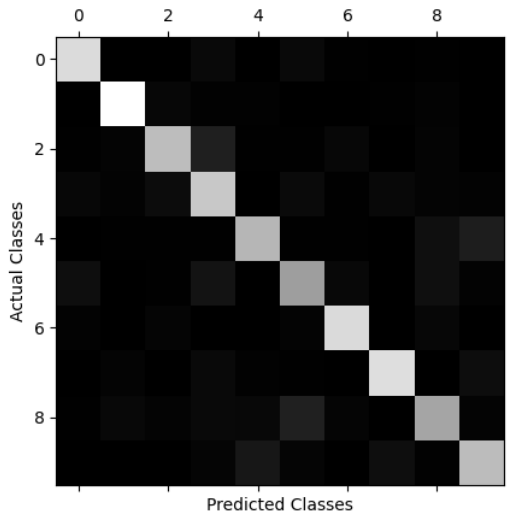
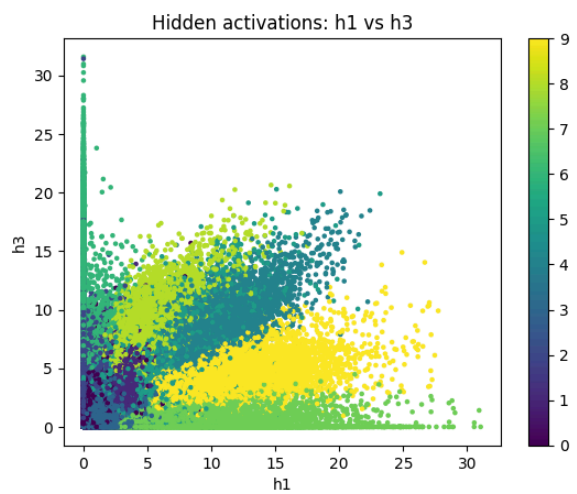
...
1719/1719 ━━━━━━━━━━━━━━━━━━━━ 53s 31ms/step - accuracy: 0.8176 - loss: 0.6376 - val_accuracy: 0.8280 - val_loss: 0.6216
313/313 ━━━━━━━━━━━━━━━━━━━━ 4s 13ms/step - accuracy: 0.7997 - loss: 0.7178
Hidden nodes: 3
Test accuracy: 0.8192
```

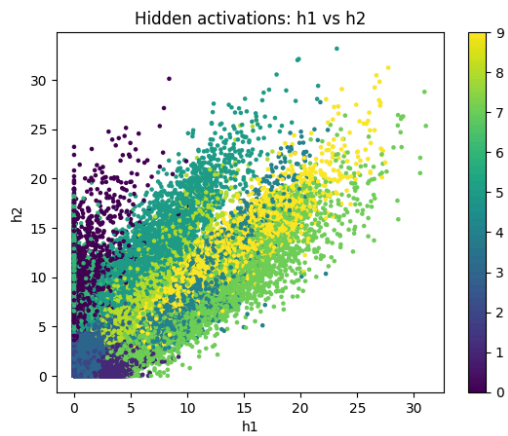


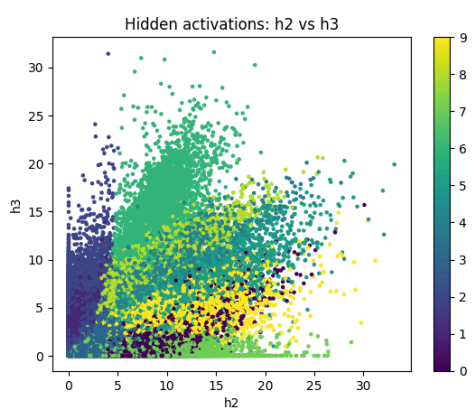






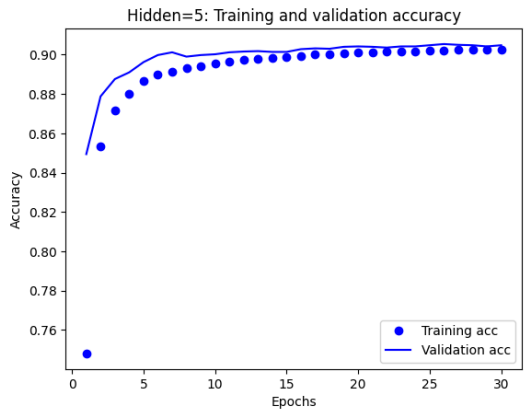


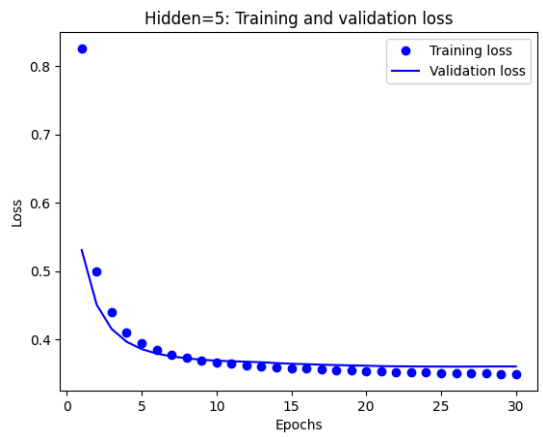


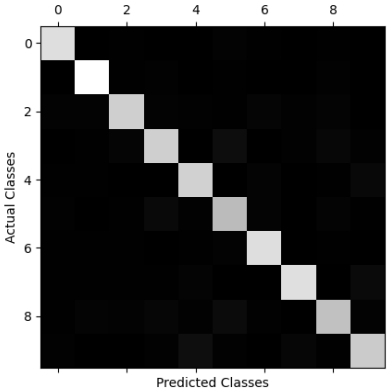


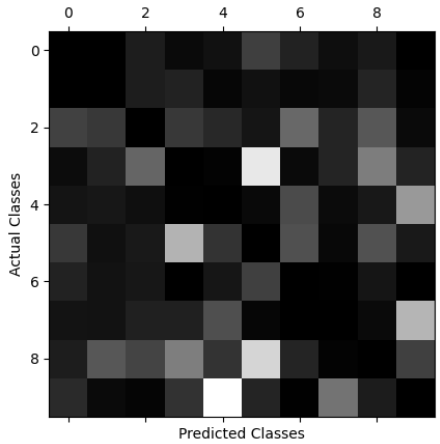
Experiment 3.2

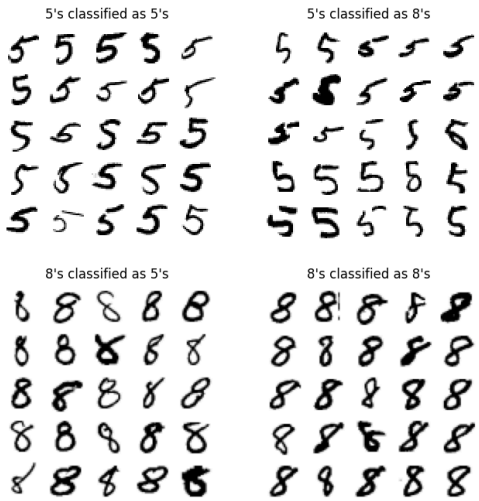
```
313/313 ━━━━━━━━━━━━━━━━━━━━ 8s 27ms/step - accuracy: 0.8824 - loss: 0.4090
Hidden nodes: 5
Test accuracy: 0.8976, Test loss: 0.3662
```

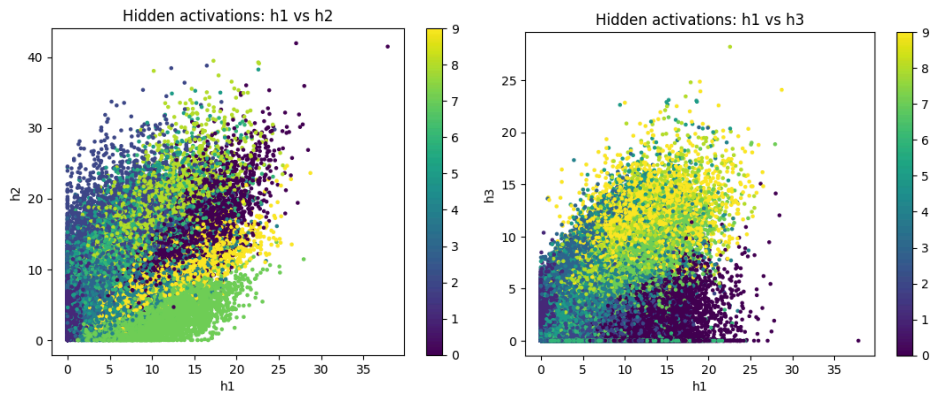


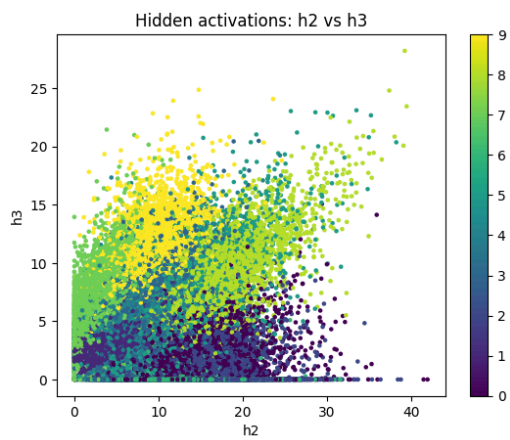



0
2
4
6
8
Actual Classes
Predicted Classes


0
2
4
6
8
Actual Classes
Predicted Classes


5's classified as 5's
5's classified as 8's
8's classified as 5's
8's classified as 8's





Experiment 3.3

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313/313 ━━━━━━━━━━━━━━━━━━━━ 8s 24ms/step - accuracy: 0.9214 - loss: 0.2886
Hidden nodes: 10
Test accuracy: 0.9328, Test loss: 0.2559
```
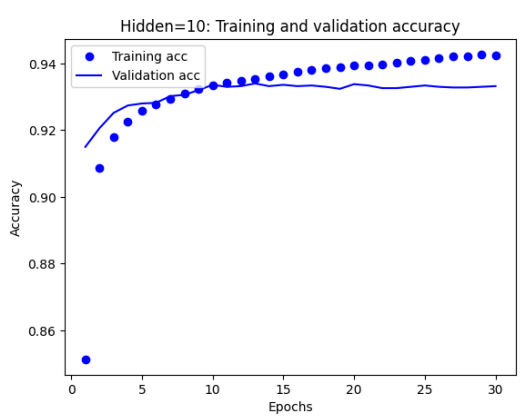
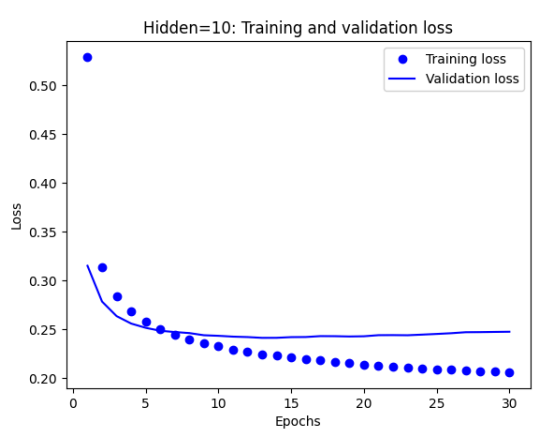
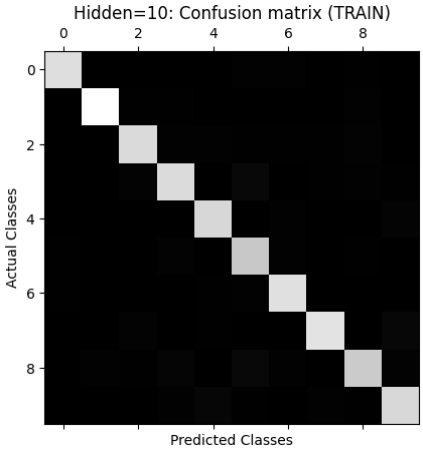
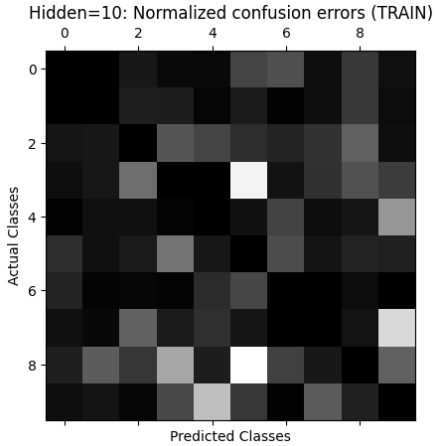
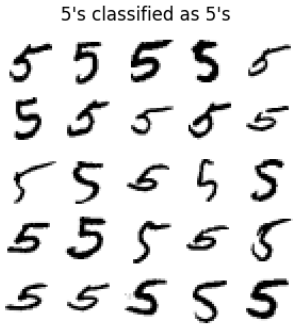
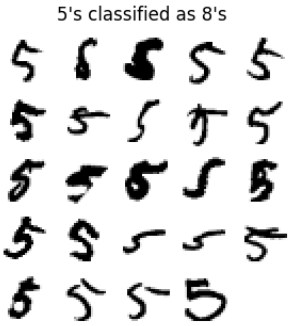



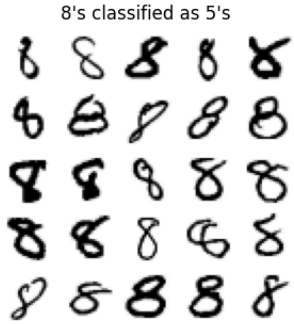


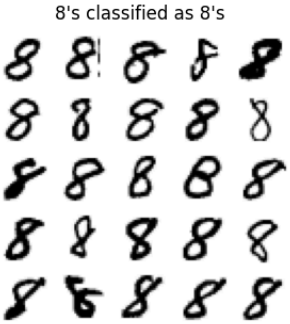

Hidden=10: Confusion matrix (TRAIN)
Actual Classes
Predicted Classes

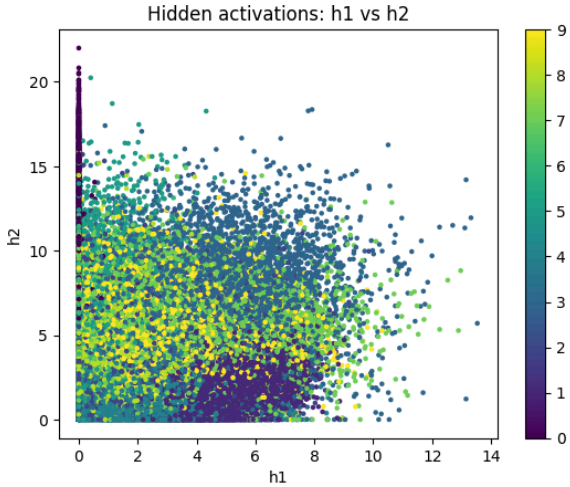

Hidden=10: Normalized confusion errors (TRAIN)
Actual Classes
Predicted Classes

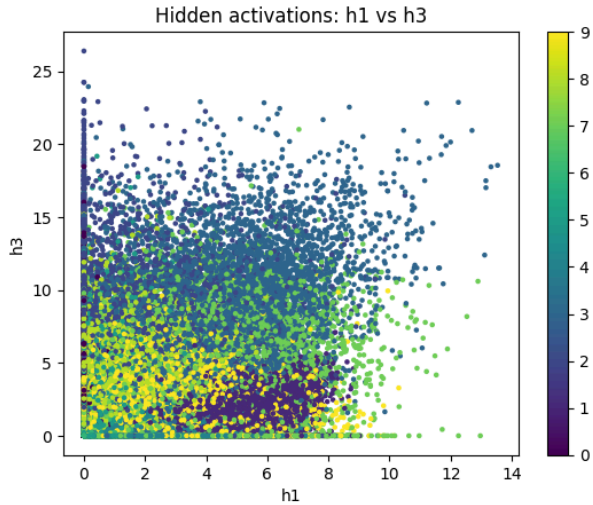

5's classified as 5's

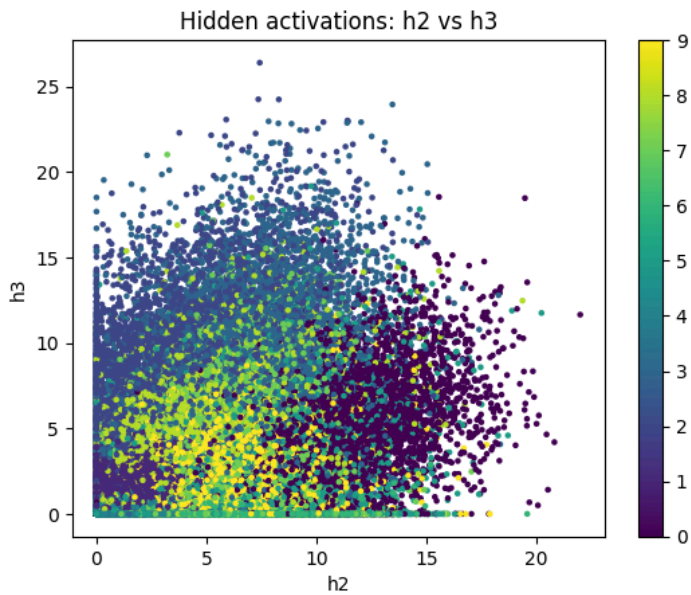

5's classified as 8's


8's classified as 5's


8's classified as 8's

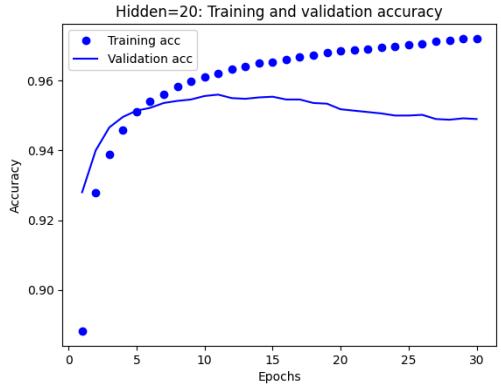


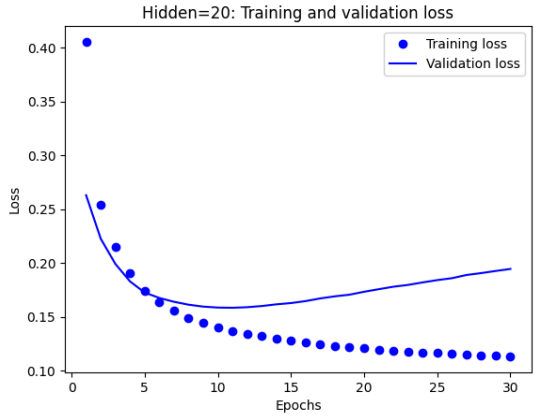


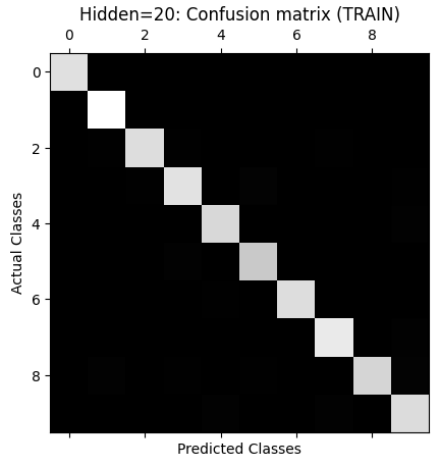


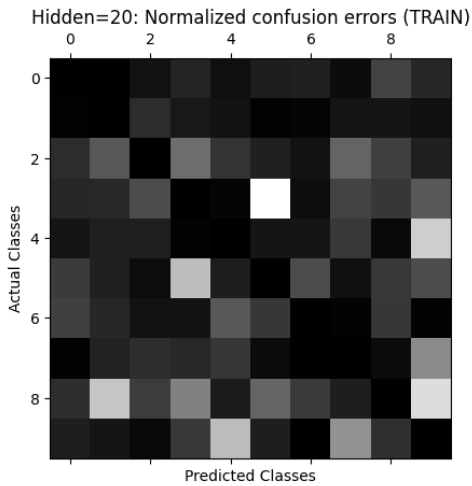
Experiment 3.4

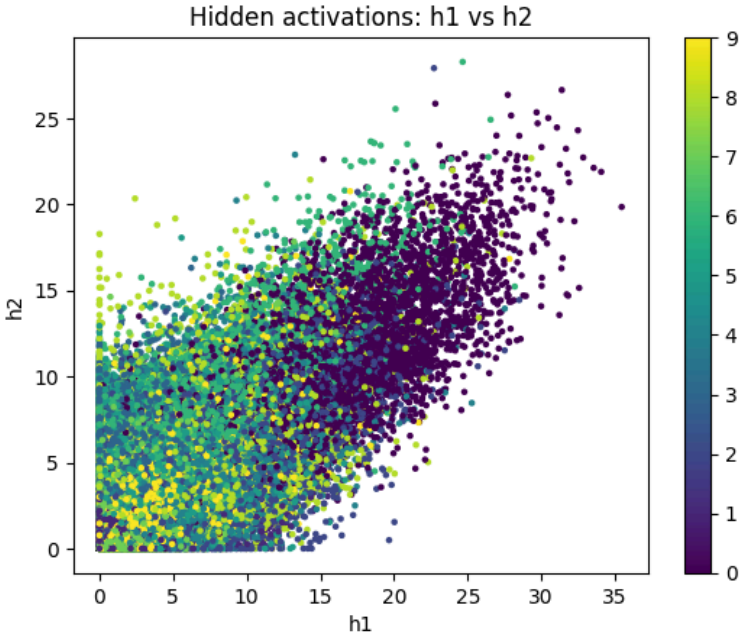
```
313/313 ━━━━━━━━━━━━━━━━━━━━ 5s 15ms/step - accuracy: 0.9442 - loss: 0.2366
Hidden nodes: 20
Test accuracy: 0.9529, Test loss: 0.2023
```

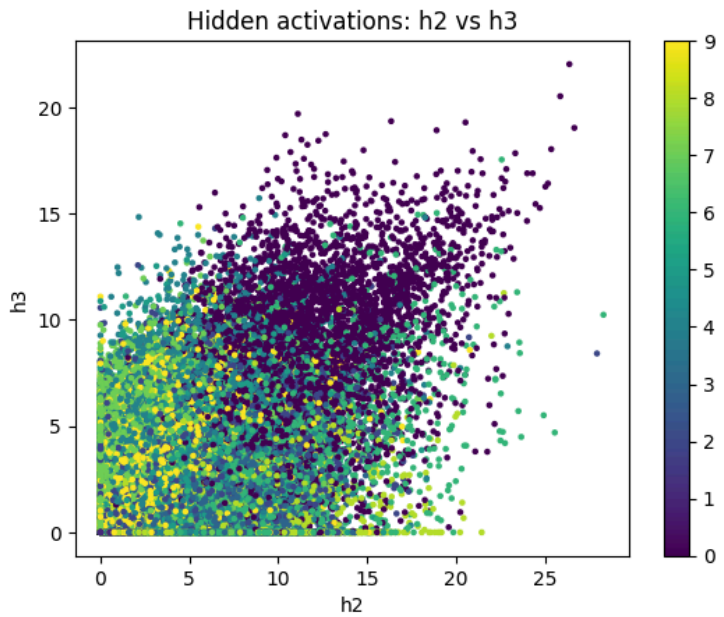

Hidden=20: Training and validation accuracy
Training acc
Validation acc
Accuracy
Epochs

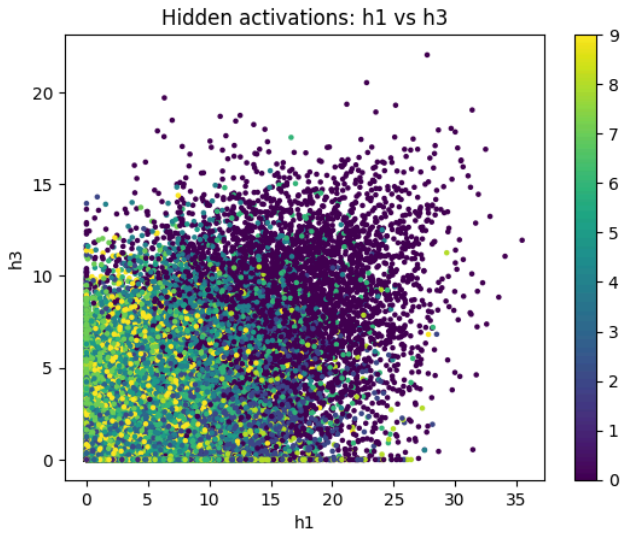

Hidden=20: Training and validation loss
Training loss
Validation loss
Loss
Epochs

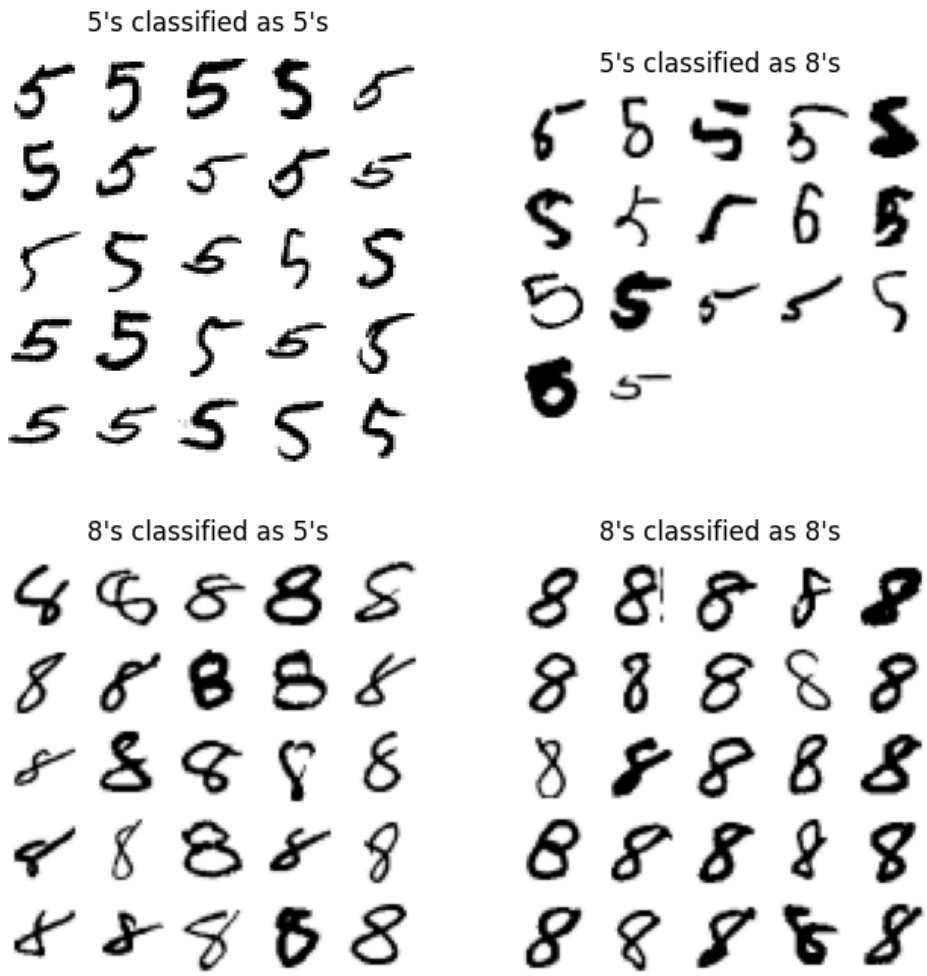

Hidden=20: Confusion matrix (TRAIN)
Actual Classes
Predicted Classes


Hidden=20: Normalized confusion errors (TRAIN)
Actual Classes
Predicted Classes

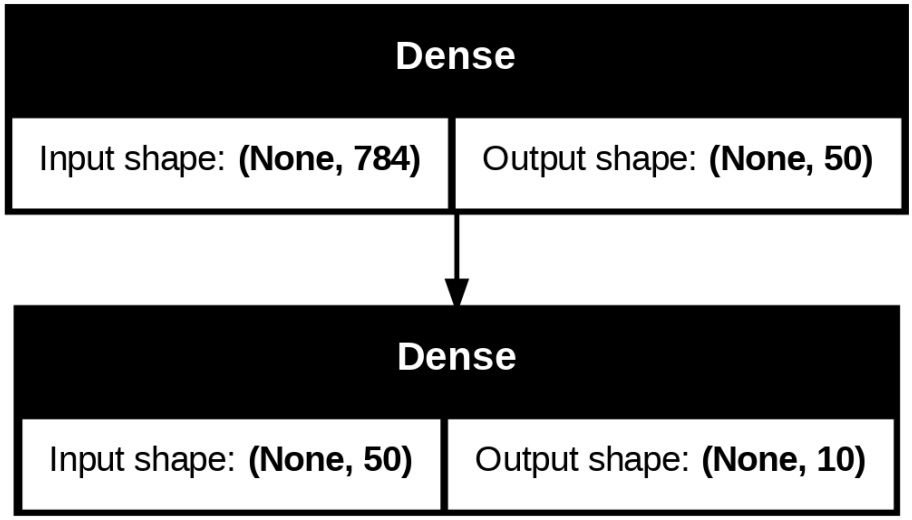

Hidden activations: h1 vs h2
h1
h2

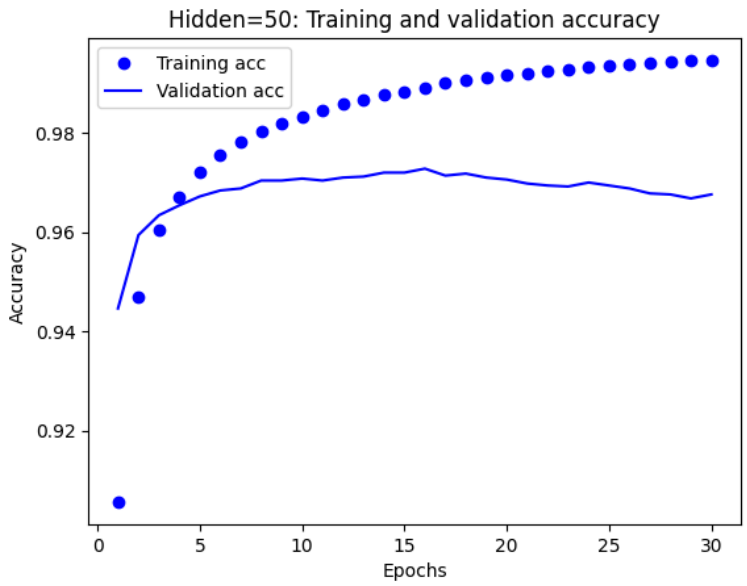

Hidden activations: h2 vs h3
h3
h2
0
5
10
15
20
25
0
5
10
15
20
0
1
2
3
4
5
6
7
8
9

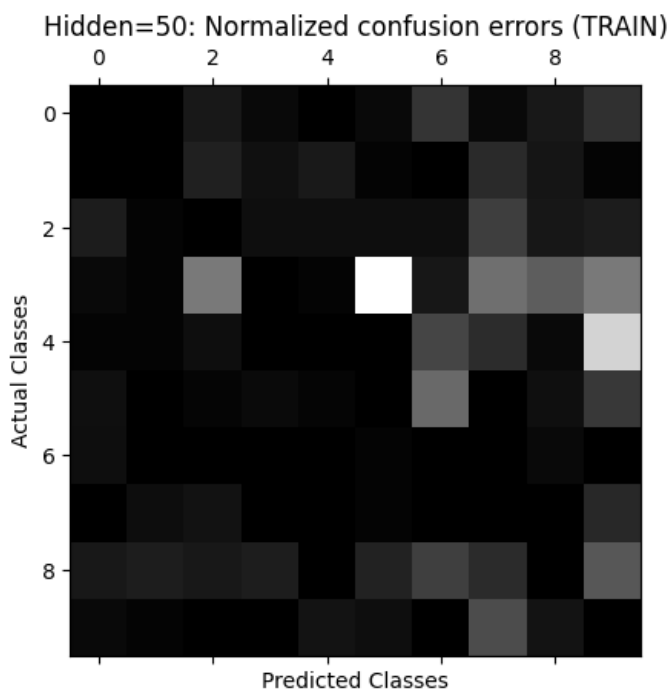


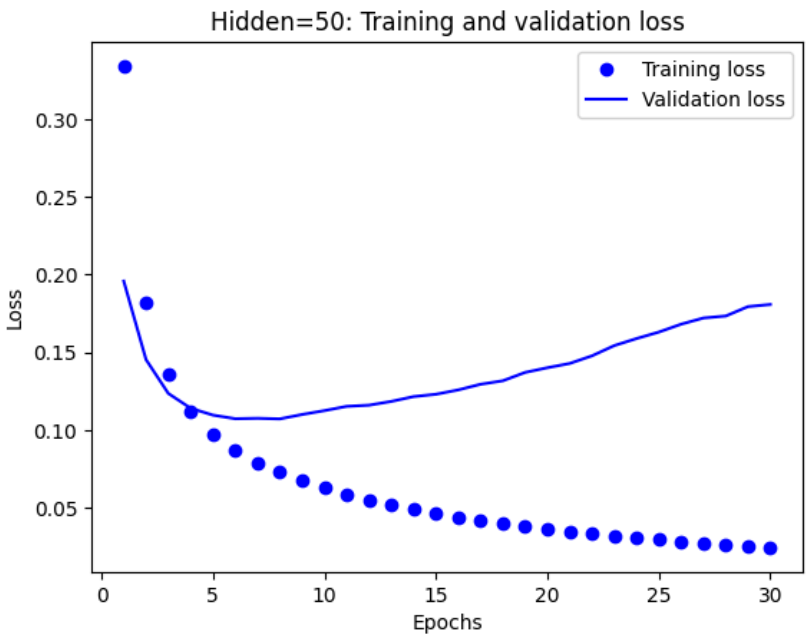


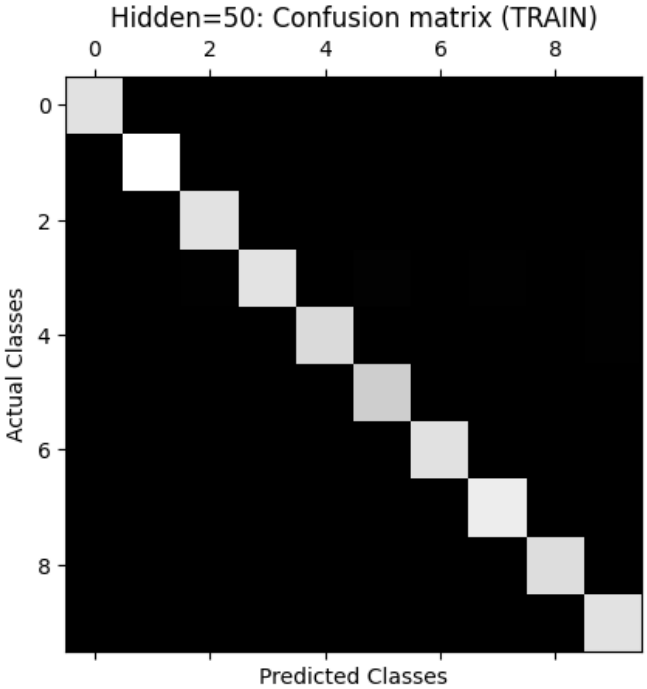
Experiment 3.5


Dense
Input shape: (None, 784)
Output shape: (None, 50)
Dense
Input shape: (None, 50)
Output shape: (None, 10)

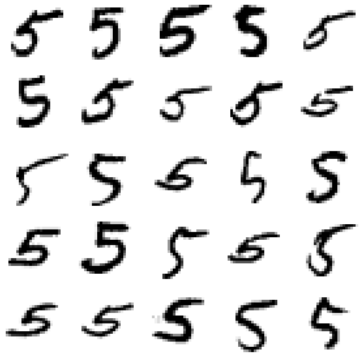

Hidden=50: Training and validation accuracy
Training acc
Validation acc
Accuracy
0.98
0.96
0.94
0.92
0
5
10
15
20
25
30
Epochs


Hidden=50: Normalized confusion errors (TRAIN)
0
2
4
6
8
Actual Classes
Predicted Classes


Hidden=50: Training and validation loss
Training loss
Validation loss
Loss
0.30
0.25
0.20
0.15
0.10
0.05
0
5
10
15
20
25
30
Epochs

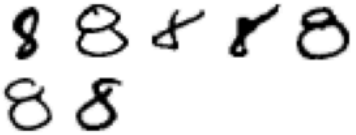


5's classified as 5's

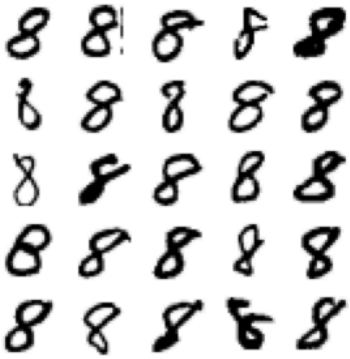


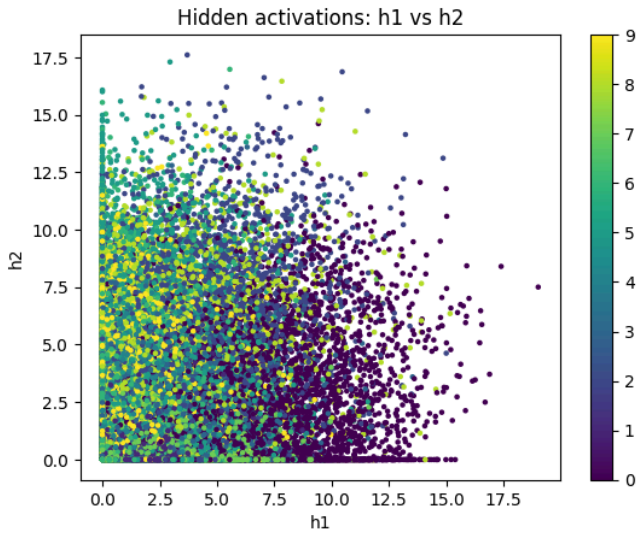
5's classified as 8's

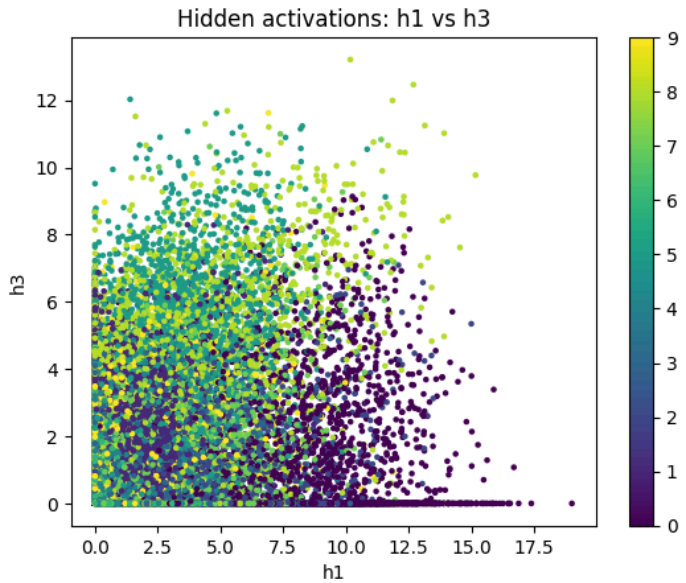
8's classified as 5's

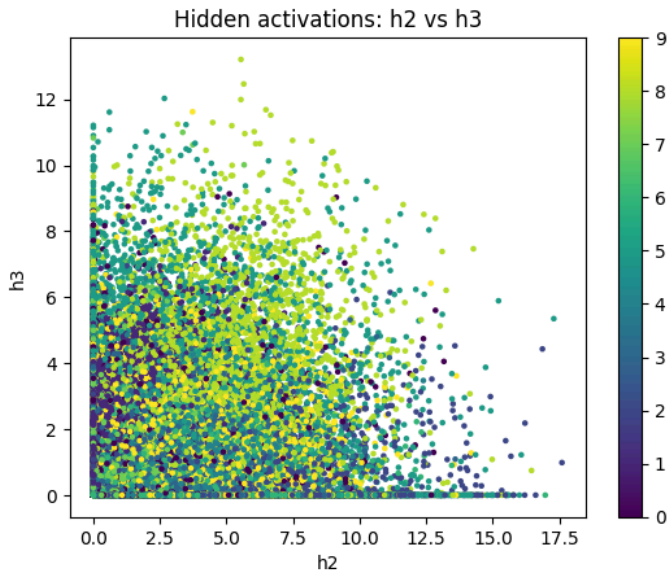


8's classified as 8's



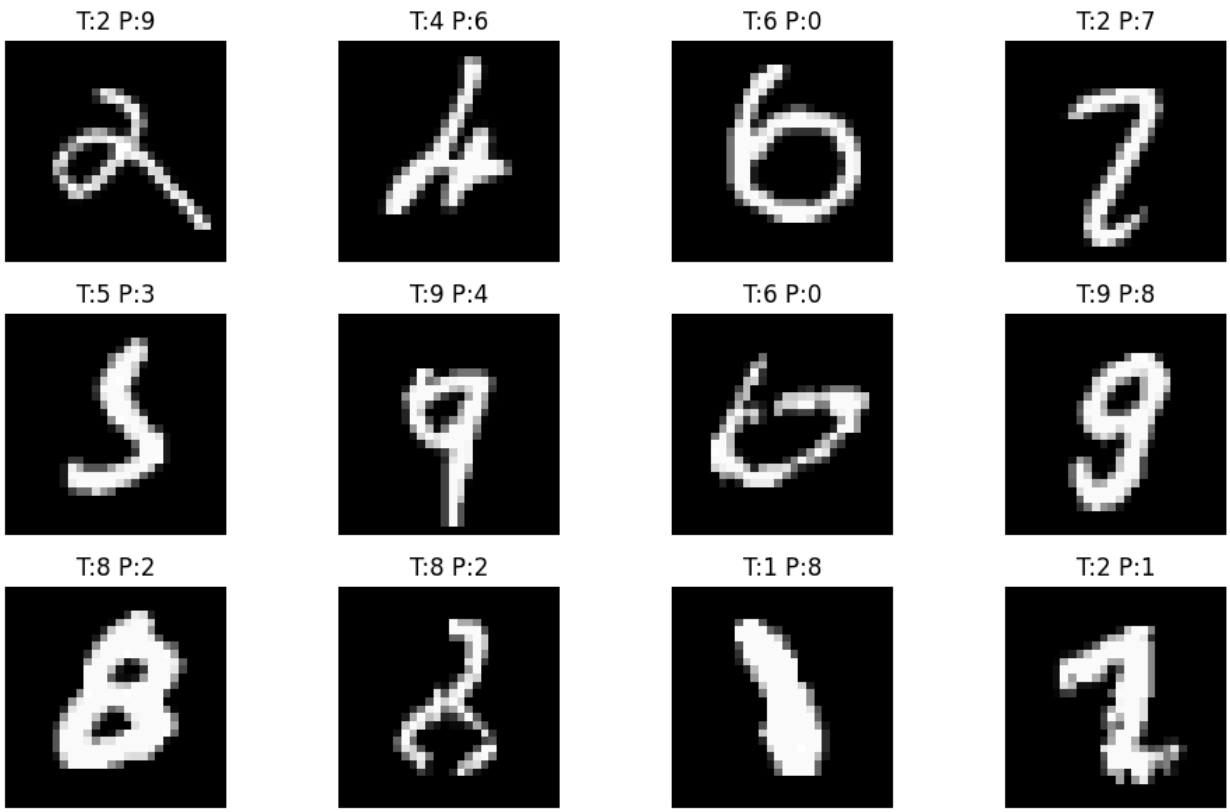

Hidden activations: h1 vs h2
h2
h1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
1
2
3
4
5
6
7
8
9

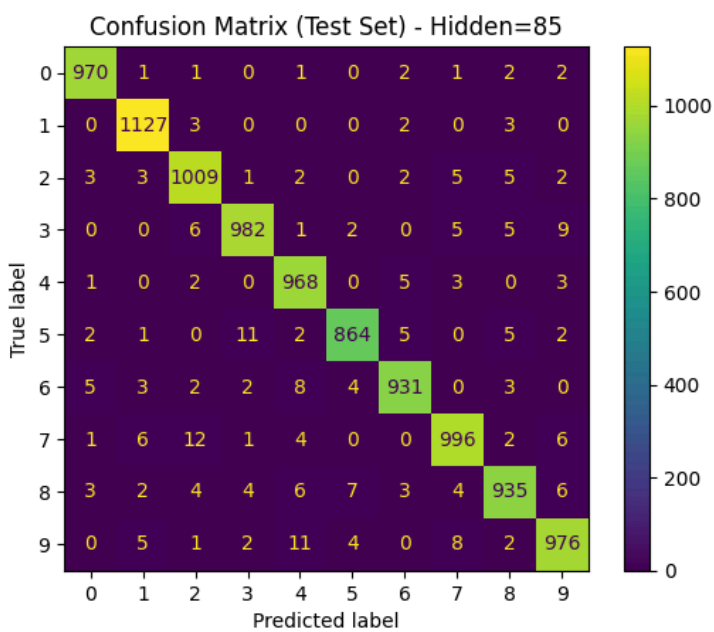

Hidden activations: h1 vs h3
h3
h1
0
2
4
6
8
10
12
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
1
2
3
4
5
6
7
8
9



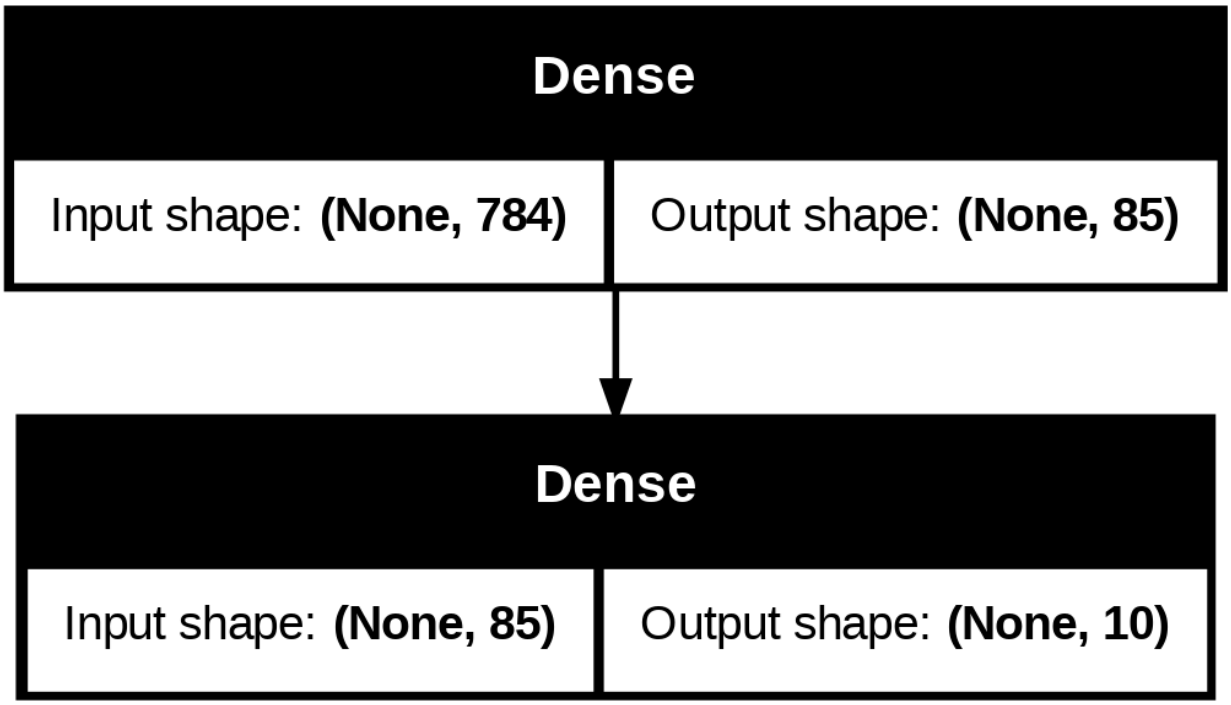
Experiment 3.6

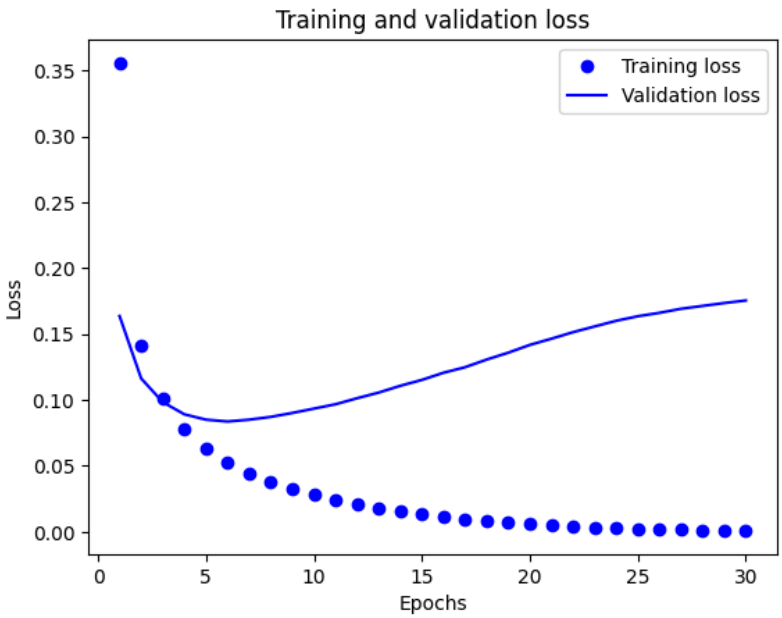
```
/usr/local/lib/python3.12/dist-packages/tensorflow/python/data/ops/structured_function.py:
  warnings.warn(
Hidden: 85 | Test acc: 0.9758 | Test loss: 0.1289
```

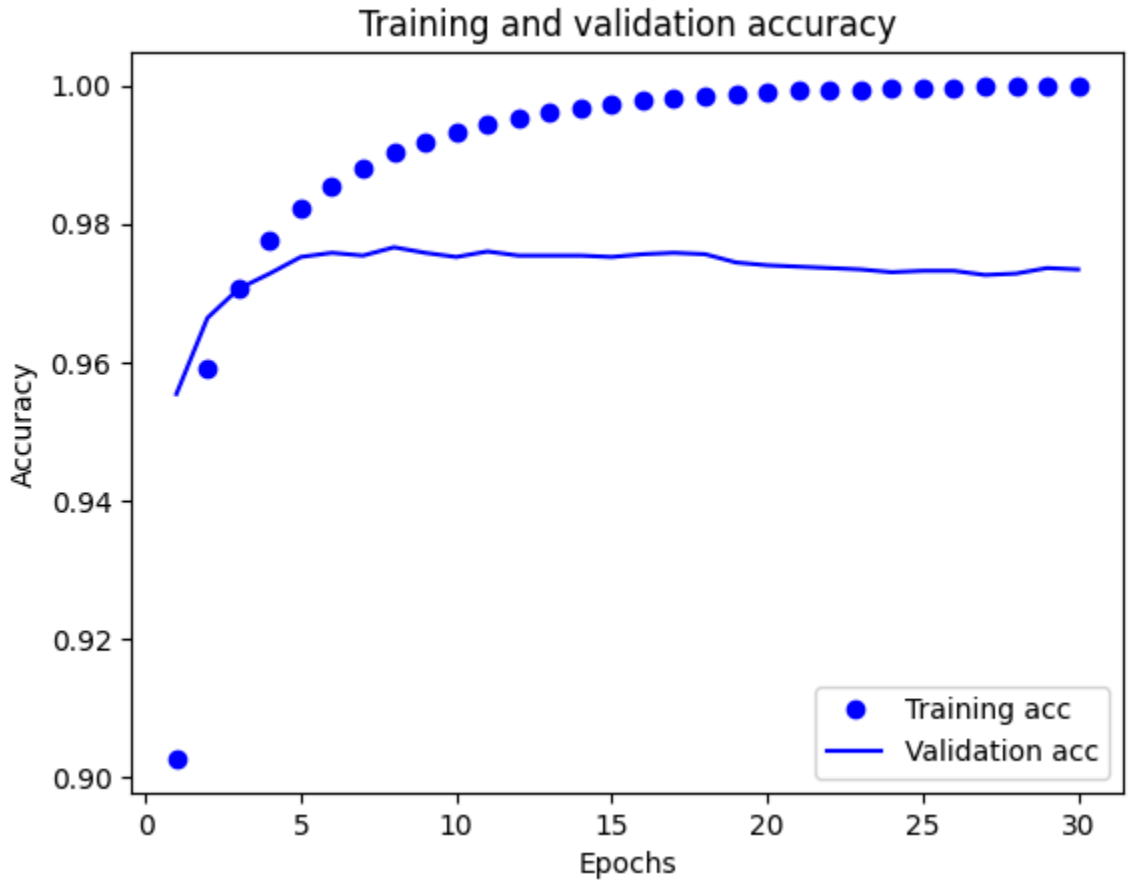




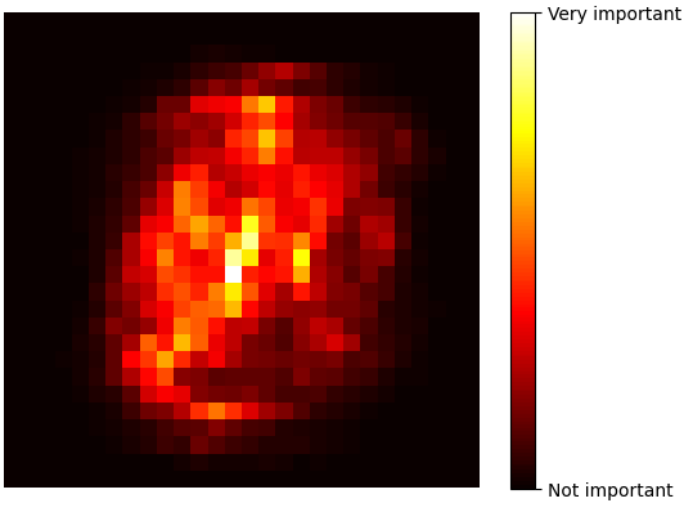
Experiment 4

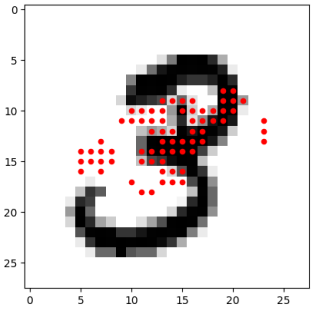

Dense
Input shape: (None, 784)
Output shape: (None, 85)
Dense
Input shape: (None, 85)
Output shape: (None, 10)

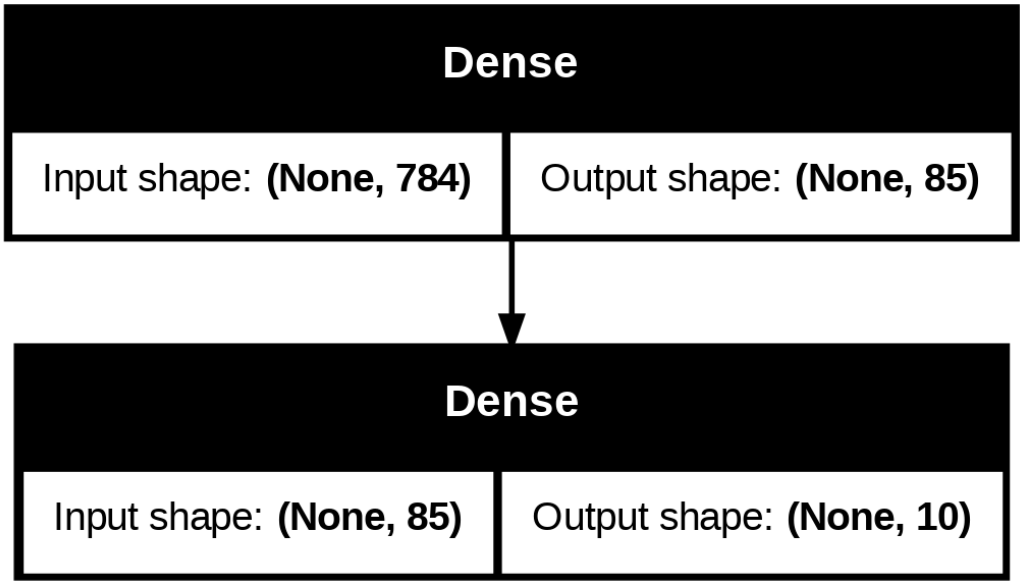


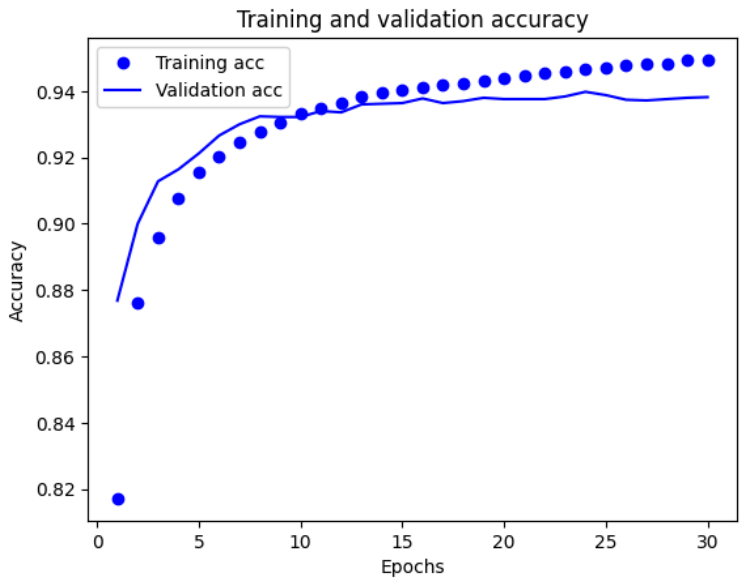


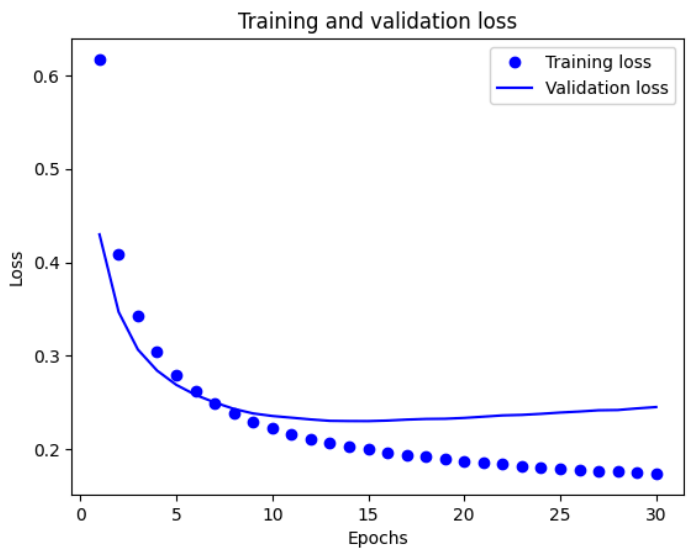
Experiment 5


Very important
Not important


0
5
10
15
20
25
0
5
10
15
20
25


Dense
Input shape: (None, 784)
Output shape: (None, 85)
Dense
Input shape: (None, 85)
Output shape: (None, 10)


Training and validation accuracy
Training acc
Validation acc
Accuracy
Epochs
0.94
0.92
0.90
0.88
0.86
0.84
0.82
0
5
10
15
20
25
30


Training and validation loss
Training loss
Validation loss
Loss
Epochs
0.6
0.5
0.4
0.3
0.2
0
5
10
15
20
25
30